



Under the hood: state, gates, and the request lifecycle

How a single run moves through consent, agents, guardrails, and persistence

A LangGraph StateGraph with a real parallel fan-out/fan-in, two conditional human-in-the-loop entry points, a deterministic authoritative verifier, and every store the pipeline touches — from the in-memory checkpointer to the two durable stores (Chroma, SQLite).

StateGraph

Fan-out/fan-in

Two HITL entry points

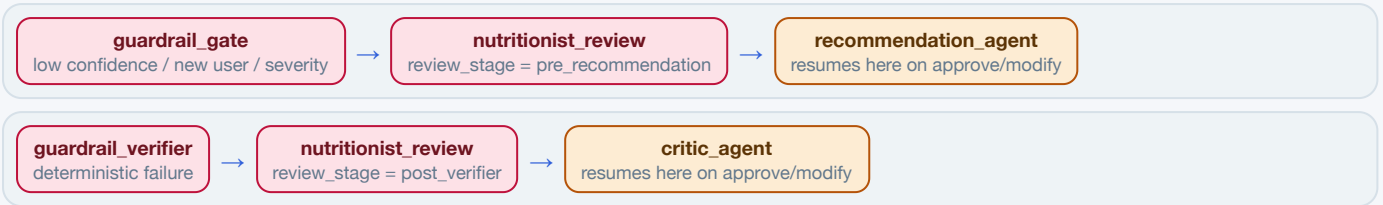
Typed fault taxonomy

Cost-tracked calls

REQUEST LIFECYCLE

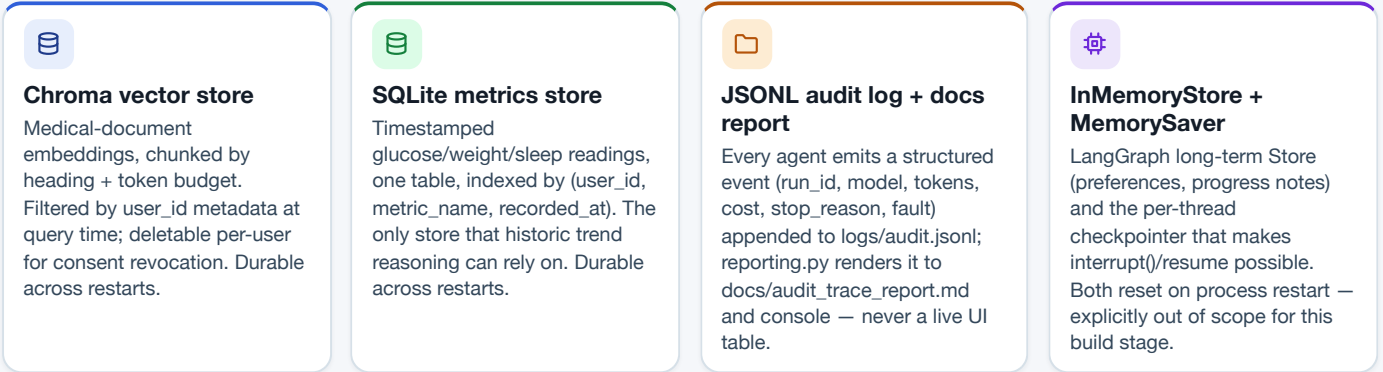
- 1 consent_gate** reads Tier-1 consent; assigns `run_id` from the graph's `thread_id`, threaded into every downstream log event.
- 2 rate_limit_gate** checks per-user image/day and analysis/day counters (in-memory, reset on restart).
- 3 vision_agent → nutrition_agent**: LLM maps food text/photo to canonical keys; a deterministic lookup table sums calories/macros — never an LLM guess.
- 4 Parallel fan-out**: `medical_rag_agent`, `activity_agent`, `sms_agent`, `calendar_agent`, and `lab_report_agent` all run concurrently off `nutrition_agent`'s output.
- 5 trend_agent** (no LLM) reads the last 90 days of glucose/weight/sleep from SQLite and classifies each trend before any model call happens.
- 6 prediction_agent** emits SAFE-category-only predictions; each is checked against an allowlist and confidence-capped before `guardrail_gate` decides HITL vs. `recommendation_agent`.
- 7 recommendation_agent → guardrail_verifier**: the deterministic verifier re-checks categories, confidence, allergens, calorie/exercise bounds, and citation ownership directly from state — independent of what any LLM concluded.
- 8 critic_agent** (advisory) only runs on a verifier-clean path; **finalize** composes the report, appends the mandatory disclaimer, and sets `run_status` to success or degraded.

TWO HUMAN-IN-THE-LOOP ENTRY POINTS

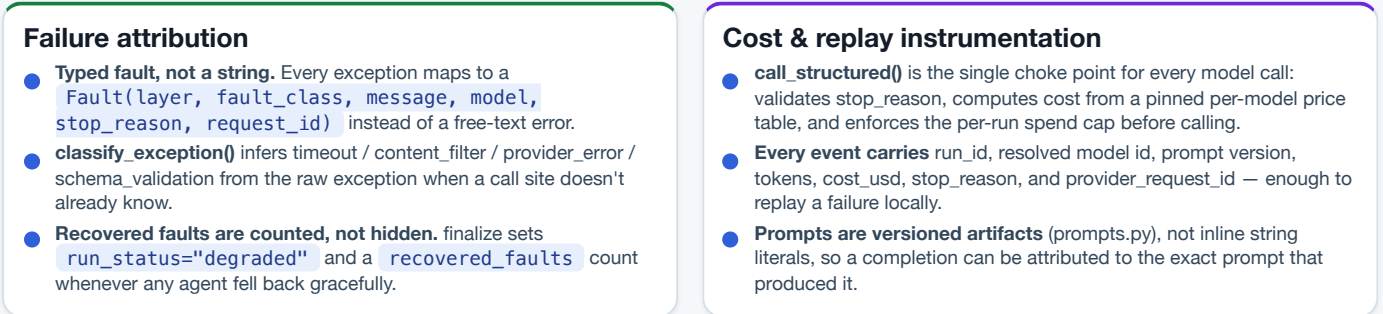


A `reject` decision from either entry point routes straight to `finalize` with no output shown to the user.

DATA STORES & PERSISTENCE



RAG RETRIEVAL FLOW



AT A GLANCE

